

# GALOIS CHARACTERISATION OF HENSELIANITY

**Definition 3.28.** Let  $(K, v)$  be a valued field and  $p$  any prime. We say  $v$  is *tamely branching* at  $p$  if

- (i)  $\text{char } Kv \neq p$ ;
- (ii)  $vK \neq pvK$ ;
- (iii)  $p^2 \mid |G_{Kv}|$ , whenever  $(vK : pvK) = p$ .

**Theorem 3.29** (Galois characterisation of henselianity). *A field  $K$  admits a henselian valuation, tamely branching at  $p$ , if and only if  $G_K$  has a Sylow  $p$ -subgroup  $P \not\cong C_2, \mathbb{Z}_p, \mathbb{Z}_2 \rtimes C_2$  that contains a non-trivial normal abelian closed subgroup  $N$ .*

recall : henselian if  $D_K = G_K$

**Remark 3.31.** (a) Let  $k$  be a field of characteristic 0 that does not admit a non-trivial henselian valuation, e.g.,  $k = \mathbb{Q}$ . Further, let  $(K, v)$  be a henselian valued field with divisible value group  $vK$  and residue field  $Kv = k$ , e.g., the field of Puiseux series over  $k$  (see Example 1.6). Then  $G_K \cong G_k$  by Facts 2.21 and 2.22. By our assumption on  $k$ , we cannot expect to recover  $v$  from the isomorphism class of  $G_K$ . Therefore, we require  $vK$  to be non- $p$ -divisible for at least one prime  $p$ , motivating (ii) in Definition 3.28.

$$K = \bigcup_{n \geq 0} k((t^{1/n})), \quad Kv = k$$

$$G_K \cong G_k \quad \text{because } R_K = \{ \text{id} \}$$

$$G_K = D_K \cong \text{Aut}(k^{\text{sep}}/k) \cong G_k$$

henselian

| Galois group                               | field                   | value group                     | residue field<br>(char. $p > 0$ ) |
|--------------------------------------------|-------------------------|---------------------------------|-----------------------------------|
| 1                                          | $L$                     | $wL$                            | $Lw$                              |
| $\in \text{Syl}_p(I)$                      | purely wild             | $p$ -divisible hull             | purely insep.                     |
| $R$                                        | $L^R$                   | $\frac{1}{p^\infty} vK \cap wL$ | $Lw \cap (Kv)^{\text{sep}}$       |
| $\text{Hom}(wL/vK, \mu_{Lw})$              | totally tamely ramified | prime-to- $p$ divisible hull    | $\parallel$                       |
| $I$                                        | $L^I$                   | $vK$                            | $Lw \cap (Kv)^{\text{sep}}$       |
| $\text{Gal}(Lw \cap (Kv)^{\text{sep}}/Kv)$ | unramified              | $\parallel$                     | rel. separable closure            |
| $D$                                        | $L^D$                   | $vK$                            | $Kv$                              |
| $G$                                        | $K$                     | $vK$                            | $Kv$                              |

(b) Consider  $K = \mathbb{C}((t))$  with the  $t$ -adic valuation. We saw in Example 2.28 that  $G_K \cong G_{\mathbb{F}_q}$  for any prime  $q$ . Therefore, we cannot expect to recover the  $t$ -adic valuation from the isomorphism class of  $G_K$ . In general, if  $\text{char } Kv \neq p$ ,  $(vK : pvK) = p$ , and  $p \nmid |G_{Kv}|$ , then  $G_K$  has Sylow  $p$ -subgroup  $\mathbb{Z}_p$ . However, this is the absolute Galois group of  $\bigcup_{p \nmid n} \mathbb{F}_{q^n}$ , which admits only the trivial valuation, hence motivating (iii).

$$G_{\mathbb{C}((t))} = D_{\mathbb{C}((t))} \cong I_{\mathbb{C}((t))} \cong \widehat{\mathbb{Z}} \cong G_{\mathbb{F}_q}$$

$\text{Aut}(\mathbb{C}^{\text{sep}}/\mathbb{C}) = \{1\}$

**Theorem 2.27.** Let  $L = K^{\text{sep}}$ . There is a semi-direct product decomposition

$$D_K/R_K \cong \text{Hom}(wL/vK, \mu_{Lw}) \rtimes G_{Kv},$$

where  $G_{Kv}$  acts naturally on the left. Moreover,

$$\text{Hom}(wL/vK, \mu_{Lw}) \cong \prod_{q \neq p} \mathbb{Z}_q^{r_q},$$

where  $q$  runs over primes away from  $p = \text{char } Kv$ , and

$$r_q := \dim_{\mathbb{F}_q}(vK/qvK)$$

denotes the  $q$ -rank of  $vK$ .

(c) If  $p > 2$ , then  $p \mid G_{Kv}$  implies  $p^2 \mid G_{Kv}$  by the Artin-Schreier Theorem 1.1. There is yet another issue if  $p = 2$  and  $2 \mid G_{Kv}$ , but  $4 \nmid G_{Kv}$ . The field  $K = \mathbb{R}((t))$ , endowed with the  $t$ -adic valuation  $v$ , is henselian with  $\text{char } Kv \neq 2$ ,  $(vK : 2vK) = 2$ , and  $G_{Kv} \cong C_2$ . By Theorem 2.27, its absolute Galois group is  $\widehat{\mathbb{Z}} \rtimes C_2$  (which has Sylow 2-subgroup  $\mathbb{Z}_2 \rtimes C_2$ ). However, it is possible to construct a field  $L$  that does not admit a non-trivial henselian valuation with  $G_L \cong \mathbb{Z}_2 \rtimes C_2$  as follows. Let  $L$  be the intersection of two distinct real closures of  $\mathbb{Q}$  in  $\mathbb{Q}^{\text{alg}}$ , and let  $\sigma$  and  $\tau$  be the involutions in  $G_{\mathbb{Q}}$  corresponding to the two real closures; together they topologically generate  $G_L$ . By Artin-Schreier, they do not commute and  $\sigma\tau$  generates an infinite cyclic group  $Z$ , where  $\sigma$  acts on  $Z$  by conjugation via  $\sigma\tau \mapsto \tau\sigma = (\sigma\tau)^{-1}$ . Hence,  $G_L = \langle \sigma, \tau \rangle$  is isomorphic to  $\overline{\mathbb{Z}} \rtimes \langle \sigma \rangle$ . If we choose the two real closures such that  $\sigma$  and  $\tau$  lie in the same Sylow 2-subgroup of  $G_{\mathbb{Q}}$ , then  $G_L$  will be of the form  $\mathbb{Z}_2 \rtimes C_2$ .

$Kv = \mathbb{R}$

**Theorem 1.1** (Artin-Schreier 1927, [AS27a, AS27b]). Let  $K$  be a field. The absolute Galois group  $G_K$  is finite if and only if

- (i)  $|G_K| = 1$ , in which case  $K$  is separably closed, or
- (ii)  $|G_K| = 2$ , in which case  $K$  is real closed, i.e., a field that admits an ordering which does not extend to any proper algebraic extension of  $K$ . Equivalently,  $K$  is real closed iff no quadratic sum  $a_1^2 + a_2^2 + \dots + a_n^2$  in  $K$  is equal to  $-1$ , any polynomial of odd degree has a root, and, for any  $a \in K$ , there exists  $b \in K$  such that  $a = b^2$  or  $a = -b^2$ .

**Theorem 3.29** (Galois characterisation of henselianity). A field  $K$  admits a henselian valuation, tamely branching at  $p$ , if and only if  $G_K$  has a Sylow  $p$ -subgroup  $P \not\cong C_2, \mathbb{Z}_p, \mathbb{Z}_2 \rtimes C_2$  that contains a non-trivial normal abelian closed subgroup  $N$ .

proof

( $\Leftarrow$ ) with henseling down (see later)  
one can reduce to

**Proposition 3.33** (Classical case). Let  $K$  be a field with  $G_K \cong \mathbb{Z}_p \rtimes \mathbb{Z}_p$ . Assume additionally that  $\sqrt{-1} \in K$  if  $p = 2$ . Then  $K$  admits a henselian valuation tamely branching at  $p$ .

key lemma

**Lemma 3.34.** Let  $K$  be a field with  $G_K \cong \mathbb{Z}_p \rtimes \mathbb{Z}_p$ . Then for every finite extension  $L/K$ , also  $G_L \cong \mathbb{Z}_p \rtimes \mathbb{Z}_p$  (with possibly different action in the semi-direct product).

Proof. Let

$$1 \longrightarrow \mathbb{Z}_p \xrightarrow{\iota} G_K \xrightarrow{\pi} \mathbb{Z}_p \longrightarrow 1$$

be the short exact sequence associated to the semidirect product for  $G_K$ . As  $L/K$  is a finite extension,  $G_L$  is a closed subgroup of finite index in  $G_K$ . Hence,  $\iota^{-1}(G_L)$  and  $\pi(G_L)$  will also be closed of finite index in their respective copies of  $\mathbb{Z}_p$ . Therefore,  $\iota^{-1}(G_L)$  and  $\pi(G_L)$  are themselves isomorphic to  $\mathbb{Z}_p$ . The associated short sequence with maps  $\iota|_{\iota^{-1}(G_L)}$  and  $\pi|_{G_L}$  remains exact and split, so the claim follows.  $\square$

$$1 \rightarrow \iota^{-1}(G_L) \rightarrow G_L \xrightarrow{\pi|_{G_L}} \pi(G_L) \rightarrow 1$$

Recall that an element  $x \in K^\times \setminus K^{\times p}$  is called  $p$ -rigid with respect to  $K^{\times p}$  if

$$K^{\times p} + xK^{\times p} \subseteq \bigcup_{i=0}^{p-1} x^i K^{\times p}. \quad (3.11)$$

If all such  $x$  are  $p$ -rigid and  $(K^\times : K^{\times p}) > p$ , then Theorem 2.69 allows us to create a valuation for the choice of parameters  $S = T_S = K^{\times p}$ . The key idea of the proof is that the right-hand side of (3.11) is given by the image of the norm map for  $K(\sqrt[p]{x})/K$ .

We organise the proof into six steps.

- Step 1.** Show  $\text{char } K \neq p$  and  $\zeta_p \in K$ . Deduce  $\dim_{\mathbb{F}_p} K^\times / K^{\times p} = 2$ .
- Step 2.** Show that every  $x \in K \setminus K^{\times p}$  is  $p$ -rigid via Kummer theory.
- Step 3.** Create a valuation  $v$  satisfying  $vK \neq pvK$  using Theorem 2.69.
- Step 4.** Show that  $\text{char } Kv \neq p$ .
- Step 5.** Show that  $v$  is henselian.
- Step 6.** Show that  $v$  is tamely branching at  $p$ .

**Step 1.** Note that  $K$  admits precisely two linearly disjoint  $C_p$ -extensions because  $\mathbb{Z}_p \rtimes \mathbb{Z}_p$  has rank two. Suppose for now that  $\text{char } K = p$ . Then, by Artin-Schreier theory (Theorem 3.24),

$$\dim_{\mathbb{F}_p} K/\wp(K) = 2.$$

Let  $a\wp(K)$  and  $b\wp(K)$  be a basis for  $K/\wp(K)$ . The splitting fields of  $X^p - X - a$  and  $X^p - X - b$  are then linearly disjoint. If  $x$  is a root of  $X^p - X - a$ , then  $\alpha := -x^{p-1}$  satisfies  $\text{Tr}(\alpha) = 1$  (as in (3.7) in the proof of Lemma 3.25), where  $\text{Tr} = \text{Tr}_{K(x)/K}$ .

**Theorem 3.24** (Artin-Schreier theory). Let  $K$  be a field of positive characteristic  $p$ . Then there exists a bijection

$$\{\text{one-dim. subspaces } \langle c \rangle + \wp(K) \leq K/\wp(K)\} \longleftrightarrow \{L/K \text{ Galois extensions of deg. } p\}$$

$$\langle c \rangle + \wp(K) \longmapsto K(\alpha),$$

where  $\wp(X) = X^p - X$  is the Artin-Schreier map and  $\alpha$  is a root of the Artin-Schreier polynomial  $\wp(X) - c$ .

$$\downarrow$$

$$x^p - x - a = 0$$

$$\Rightarrow -x^{p-1} = -1 - ax^{-1}$$

$$\begin{aligned} \text{Tr}(\alpha) &= \text{Tr}(-x^{p-1}) = \text{Tr}(-1 - ax^{-1}) = -\text{Tr}(1) - a\text{Tr}(x^{-1}) \\ &= -p - a\text{Tr}(x^{-1}) \\ &= -p + 1 \end{aligned}$$

$$x^p - x - a \xrightarrow[\text{by } -x^p a]{\text{divide up}} -a^{-1} + a^{-1}(x^{-1})^{p-1} + (x^{-1})^p = 0$$

$$\Rightarrow \text{Tr}_{\frac{1}{K}}(x^{-1}) = -a^{-1}$$

We claim that the elements  $a\alpha$ ,  $b\alpha$ , and  $b$  are  $\mathbb{F}_p$ -linearly independent in  $K(x)/\wp(K(x))$ . Indeed, let  $i_1, i_2, i_3 \in \mathbb{F}_p$  be such that

$$i_1 a\alpha + i_2 b\alpha + i_3 b = y^p - y$$

for some  $y \in K(x)$ . Taking the trace on both sides of the equation, we obtain

$$i_1 a + i_2 b + p(i_3 b) = i_1 a + i_2 b = \text{Tr}(y^p) - \text{Tr}(y) = \text{Tr}(y)^p - \text{Tr}(y),$$

which implies  $i_1 = i_2 = 0$  by our choice of  $a$  and  $b$ . Hence,  $y^p - y - i_3 b = 0$ . By our choice of  $b$ , this implies  $i_3 = 0$ , and therefore,  $\dim_{\mathbb{F}_p} K(x)/\wp(K(x)) \geq 3$ . Recall that Lemma 3.34 tells us that  $G_{K(x)} \cong \mathbb{Z}_p \rtimes \mathbb{Z}_p$ , which contradicts our initial observation applied to  $K(x)$ . Therefore,  $\text{char } K \neq p$  and  $\zeta_p \in K^{\text{sep}}$ .

Note that  $[K(\zeta_p) : K] \leq p - 1$  and the assumption that  $G_K$  is a pro- $p$  group imply  $\zeta_p \in K$ . In particular,  $\dim_{\mathbb{F}_p} K^\times / K^{\times p} = 2$  by Kummer theory (Theorem 3.23).

**Theorem 3.23** (Kummer theory). Let  $K$  be a field with  $\text{char } K \neq p$  and  $\zeta_p \in K$ . Then there exists a bijection

Key Lemma

$$\begin{aligned} \{\text{one-dim. subspaces } \langle c \rangle K^{\times p} \leq K^\times / K^{\times p}\} &\longleftrightarrow \{L/K \text{ Galois extensions of deg. } p\} \\ \langle c \rangle K^{\times p} &\longmapsto K(\sqrt[p]{c}) \\ (K^\times \cap L^{\times p}) / K^{\times p} &\hookrightarrow L. \end{aligned}$$

Step 2

Claim: Every  $x \in K^\times \setminus K^{\times p}$  is  $p$ -rigid

Recall that an element  $x \in K^\times \setminus K^{\times p}$  is called  $p$ -rigid with respect to  $K^{\times p}$  if

$$K^{\times p} + xK^{\times p} \subseteq \bigcup_{i=0}^{p-1} x^i K^{\times p}. \quad (3.11)$$

**Step 2.** Let  $x \in K^\times \setminus K^{\times p}$  and set  $L := K(\sqrt[p]{x})$ . Choose two cosets  $xK^{\times p}, yK^{\times p}$  that form an  $\mathbb{F}_p$ -basis for  $K^\times / K^{\times p}$ . By Kummer theory,  $\sqrt[p]{x}L^{\times p}$  and  $yL^{\times p}$  are linearly independent in  $L^\times / L^{\times p}$  and hence form a basis by Lemma 3.34. Thus, we may write

$$L^\times = \bigcup_{0 \leq i, j < p} (\sqrt[p]{x})^i y^j L^{\times p}.$$

For any  $a, b \in K^\times$ , there exist  $i, j \in \{0, \dots, p-1\}$  and  $z \in L^\times$  such that

$$a + b\sqrt[p]{x} = (\sqrt[p]{x})^i y^j z^p.$$

Note that  $a + b\sqrt[p]{x}$  has minimal polynomial  $(X - a)^p - b^p x$ . Applying the norm map to both sides yields

$$(-1)^{p-1} N_{L/K}(a + b\sqrt[p]{x}) = (-1)^{p-1} a^p + x b^p = x^i y^{jp} N_{L/K}(z)^p \in x^i K^{\times p}.$$

This proves the inclusion (3.11):  $x$  is  $p$ -rigid. (Note that we assumed  $\sqrt{-1} \in K$  if  $p = 2$ .)

**Step 3.** By Step 2, any  $x \in K^\times \setminus K^{\times p}$  is  $p$ -rigid. Moreover,  $(K^\times : K^{\times p}) = p^2 > p$ . By the creation of valuations from  $p$ -rigid elements (Theorem 2.69) for parameters  $S = T_S = K^{\times p}$ , we obtain a non-trivial valuation  $v$  with  $vK \neq pvK$ . Additionally, we replace  $v$  by the coarsening with respect to the maximal  $p$ -divisible convex subgroup. Then,  $v$  will not contain any non-trivial  $p$ -divisible convex subgroups and  $v$  remains non- $p$ -divisible.

**Step 4.** We now prove  $\text{char } Kv \neq p$ . Assume towards a contradiction that  $\text{char } Kv = p$ . Then,  $\text{Conv}(vp)$  is a non-trivial convex subgroup. By our choice of  $v$ , we know that  $\text{Conv}(vp)$  and hence  $(0, vp] \subset vK$  is not  $p$ -divisible. So we may choose an element  $x \in K$  such that

$$0 < vx \leq vp \quad \text{and} \quad vx \notin pvK.$$

In particular,  $x \in K^\times \setminus K^{\times p}$  is  $p$ -rigid, so there exist  $i \in \{0, \dots, p-1\}$  and  $y \in K^\times$  such that  $1 + x = x^i y^p$ . Note that  $v(1 + x) = 0$  implies

$$i = 0, \quad vy = 0, \quad \text{and} \quad y \equiv 1 \pmod{\mathcal{M}_v},$$

the latter as in (3.5). If we write  $y = 1 + z$  with  $z \in \mathcal{M}_v$ , then

$$vx = v((1 + z)^p - 1) \geq \min\{vp + vz, v(z^p)\}.$$

But  $vx < vp + vz$  implies  $vx = v(z^p)$ , and hence  $vx \in pvK$ , contrary to our choice of  $x$ .

**Step 5.** Assume that  $v$  is not henselian. By Lemma 2.37, there exists a Galois extension  $L/K$  of degree  $p$  with pairwise inequivalent prolongations  $v_1, \dots, v_g$  of  $v$  to  $L$  with  $g > 1$ . By the fundamental equality (Corollary 2.25),  $p = \text{def } g$  and therefore

$$d = e = f = 1 \quad \text{and} \quad g = p,$$

where  $v_1, \dots, v_p$  are thus immediate, incomparable prolongations (cf. Fact 2.12). By Kummer theory,  $L = K(\sqrt[p]{x})$  for some  $x \in \mathcal{O}_v^\times$  (otherwise, we would have  $e = p$ ). Set  $x_0 := \sqrt[p]{x} \notin L^{\times p}$ .

Applying the weak approximation theorem (Fact 2.13), we may find for each  $i \in \{1, \dots, p\}$ ,

$$x_i \in \mathcal{M}_{v_i} \cap \bigcap_{j \neq i} \mathcal{O}_{v_j}^\times. \quad (3.12)$$

As all  $v_i$  are immediate, we have  $v_i L = v_i K$  for each  $i$ . We may assume without loss of generality, that  $v_i x_i \notin pv_i K$ , as follows: if  $v_i x_i \in pv_i K$ , use the fact that  $\text{Conv}(v_i x_i)$  is non-trivial to find  $y \in K$  with  $v_i y \in (0, v_i x_i)$  and  $v_i y \notin pv_i K$  (otherwise, as in Step 4,  $(0, v_i x_i] \subset v_i K$  would be  $p$ -divisible, which cannot happen). Replacing  $x_i$  by  $x_i + y$ , gives the desired element satisfying (3.12) and  $v_i x_i \notin pv_i K$ .

We claim that  $L(\sqrt[p]{x_0}, \sqrt[p]{x_1}, \dots, \sqrt[p]{x_p})/L$  has Galois group  $C_p^{p+1}$ . By Kummer theory, it suffices to show that

$$x_0^{i_0} x_1^{i_1} \cdots x_p^{i_p} \in L^{\times p} \text{ with } i_0, \dots, i_p \in \{0, \dots, p-1\} \implies i_0 = \dots = i_p = 0.$$

Evaluating this product by  $v_j$  yields  $i_j = 0$  for every  $j = 1, \dots, p$ . Recall that  $x_0 \notin L^{\times p}$ , so  $i_0 = 0$  as well. By Step 1 and Lemma 3.34, we know that  $L$  admits at most two linearly disjoint  $C_p$ -extensions, so our construction is impossible. We conclude that  $v$  must be henselian.

**Definition 2.36.** A valued field  $(K, v)$  is called  *$p$ -henselian* if  $v$  extends uniquely to any finite Galois extension of  $p$ -power degree, or equivalently, to the maximal pro- $p$  Galois extension of  $K$ , which we denote by  $K(p)$ .

**Lemma 2.37** ([EP05, Thm. 4.2.2]). *Let  $(K, v)$  be a valued field. Then  $v$  is  $p$ -henselian if and only if  $v$  extends uniquely to any  $C_p$ -extension, i.e., any Galois extension of degree  $p$ .*

**Corollary 2.25** (Fundamental equality). *Let  $(L, w)/(K, v)$  be a finite Galois extension of valued fields. Let  $g$  be the number of prolongations of  $v$  to  $L$  (all of which are conjugate). Then*

$$[L : K] = \text{def } g,$$

where  $d, e, f$  are the ramification invariants of  $(L, w)/(L^D, w|_{L^D})$ .

**Fact 2.12.** *Let  $(K, v)$  be a valued field and  $L/K$  an algebraic extension. If  $w_1$  and  $w_2$  denote valuations on  $L$  extending  $v$ , then  $\mathcal{O}_{w_1} \subseteq \mathcal{O}_{w_2}$  implies  $\mathcal{O}_{w_1} = \mathcal{O}_{w_2}$ .*

**Fact 2.13** (Weak approximation theorem). *Let  $\mathcal{O}_{v_1}, \dots, \mathcal{O}_{v_n}$  be pairwise incomparable valuation rings on  $K$ . Then, for any tuple  $(x_i)_{i=1}^n \in \prod_{i=1}^n \mathcal{O}_{v_i}$ , there exists  $x \in K$  such that  $x - x_i \in \mathcal{M}_{v_i}$  for  $i = 1, \dots, n$ .*

immediate  
 $e(w/v) = f(w/v) = 1$

**Step 6.** Assume  $(vK : pvK) = p$  and let  $x \in K$  be such that  $vx \notin pvK$ . It remains to verify property (iii), that is,  $p^2 \mid |G_{Kv}|$ . If  $G_{Kv} = 1$ , then  $\mathcal{O}_v^\times = \mathcal{O}_v^{\times p}$ , since  $(Kv)^\times = (Kv)^{\times p}$  and  $1 + \mathcal{M}_v = (1 + \mathcal{M}_v)^p$  by henselianity. This would imply that there is only one  $C_p$ -extension, namely  $K(\sqrt[p]{x})$ . Hence  $G_{Kv} \neq 1$  (and we are done by the Artin-Schreier Theorem 1.1 in the case  $p \neq 2$ ).

Let  $L := K(\sqrt[p]{y})$ , where  $y \in \mathcal{O}_v^\times$  is such that  $yv \notin (Kv)^{\times p}$ . In particular,  $L/K$  is unramified of degree  $f = p$ . As before, we cannot have  $G_{Lv} = 1$ , because  $L$  admits more than one  $C_p$ -extension. This shows that  $p^2 \mid G_{Kv}$ .  $\square$

## General case

From this, we are able to deduce the general case. Our main tool will be “Henseling Down” (Theorem 2.54). In order to apply this theorem to a tamely branching valuation  $v$ , we require  $\mathcal{O}_v \in H(K)$  and moreover, that no proper coarsening of  $v$  has real closed residue field. Conveniently, we may assume this extra condition without loss of generality:

**Theorem 2.54** (Henseling Down, [Koe03, Prop. 3.1]). *Let  $L/K$  be an algebraic extension with  $L$  not separably closed. Consider a valuation  $w$  on  $L$  with  $\mathcal{O}_w \in H(L)$ . Then, the valuation ring of the restriction  $v := w|_K$  lies in  $H(K)$  in each of the following cases:*

- (i)  $L/K$  is normal.
- (ii)  $L/K$  is finite.
- (iii)  $G_L$  is a Sylow  $p$ -subgroup of  $G_K$  for some prime  $p$ . If  $p = 2$ , additionally assume that no proper coarsening of  $w$  has real closed residue field.

## recall

**Definition 2.50.** Let  $K \neq K^{\text{sep}}$  be a field. We partition the set of henselian valuation rings on  $K$  into two sets

$$H_1(K) := \{\mathcal{O}_v : \mathcal{O}_v \text{ henselian with } Kv \text{ not separably closed}\}$$

$$H_2(K) := \{\mathcal{O}_v : \mathcal{O}_v \text{ henselian with } Kv \text{ separably closed}\}.$$

**Definition 2.52** (Canonical henselian valuation). Let  $K \neq K^{\text{sep}}$  be a field. The *canonical henselian valuation*  $v_K$  (with valuation ring  $\mathcal{O}_{v_K}$ ) is defined to be

- (i) the coarsest valuation in  $H_2(K)$  if  $H_2(K) \neq \emptyset$ , and
- (ii) the finest valuation in  $H_1(K)$  otherwise.

If  $H_2(K) \neq \emptyset$ , then  $\mathcal{O}_{v_K}$  does not lie in  $H_1(K)$ . We further define:

$$H(K) := H_1(K) \cup \{\mathcal{O}_{v_K}\}$$

Note that any valuation in  $H(K)$  is comparable to any other henselian valuation and that  $\mathcal{O}_{v_K}$  is the minimum of  $H(K)$ .

$L/K$  finite

**Proof.** By Theorem 1.1,  $N$  is torsion-free unless  $p = 2$  and  $\sqrt{-1} \notin \text{Fix } N$ . We may replace  $K$  by  $K(\sqrt{-1})$  in this exceptional case by **Henseling Down** (Theorem 2.54(ii)) and Lemma 3.36, ensuring  $P \not\cong \mathbb{Z}_2$ . The fact that  $N$  remains non-trivial is due to Lemma 2.56. In particular, we may now view  $N$  as a torsion-free  $\mathbb{Z}_p$ -module.

**Lemma 3.36.** Let  $(L, w)$  be a valued field,  $w$  tamely branching at  $p$ . If  $L/K$  is an algebraic extension of fields, then  $v = w|_K$  is tamely branching at  $p$  as well.

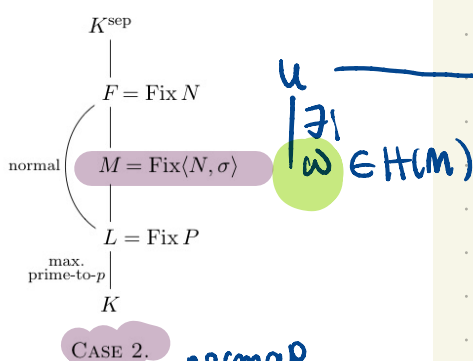
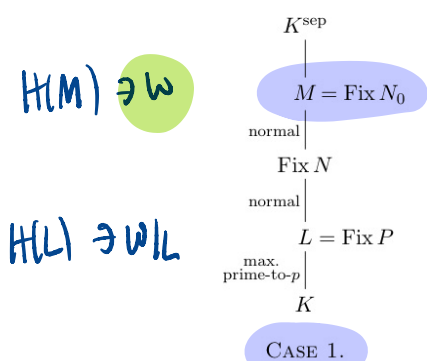
**Proof.** Recall that  $wL/vK$  is a torsion group and  $Lw/Kv$  is an algebraic extension by Fact 2.11. In particular,  $p^2 \mid G_{Lw}$  implies  $p^2 \mid G_{Kv}$ . The rest of the claim follows immediately from the following general fact about abelian groups: if  $\Delta \leq \Gamma$  are  $p$ -torsion-free abelian groups such that  $\Gamma/\Delta$  is a torsion group, then

$$\dim_{\mathbb{F}_p}(\Delta/p\Delta) \geq \dim_{\mathbb{F}_p}(\Gamma/p\Gamma).$$

See [EP05, Ex. 5.5.2] for a proof of this inequality.  $\square$

**Lemma 2.56** (cf. [Gey69, 1.11 Folg.]). Let  $F$  be a field with pro-2 absolute Galois group  $G_F$ . If  $F$  is not real closed, then  $G_F$  contains no normal subgroup isomorphic to  $C_2$ .

**Case 1.** Let us first consider the easier case that  $N$  is not topologically generated by a single element (i.e.,  $N \not\cong \mathbb{Z}_p$ ), so there exists  $N_0 < N$  with  $N_0 \cong \mathbb{Z}_p \times \mathbb{Z}_p$ . By the “classical case” (Proposition 3.33), there exists a henselian valuation  $w$  on  $M := \text{Fix } N_0$ , tamely branching at  $p$ , which we may assume to lie in  $H(M)$  by Lemma 3.35. Now, we may “hensel down” as indicated by the left diagram.



①  $\mathcal{O}_u \in H(F)$

Henseling down to  $u|_K$

②  $u, v_F \in H_2(F)$

$\mathcal{O}_u \subsetneq \mathcal{O}_{v_F}$

Henseling down to  $v_F|_K$

More precisely, two applications of **Henseling Down** (Theorem 2.54(i) and Lemma 3.36) show that  $w|_L$  is in  $H(L)$  and tamely branching at  $p$ , where  $L := \text{Fix } P$ . By Lemma 3.35, again, we may additionally assume that  $L$  admits a henselian valuation  $w'$ , tamely branching at  $p$ , such that no proper coarsening has real closed residue field if  $p = 2$ . Therefore,  $w'|_K$  is the desired valuation by **Henseling Down**, Theorem 2.54(iii). **flow**

**Case 2.** Let us now consider the case that  $N \cong \mathbb{Z}_p$ . As  $P \not\cong \mathbb{Z}_p$ , we may find some  $\sigma \in P \setminus N$  such that  $\langle \sigma \rangle \cong \mathbb{Z}_p$  and  $\langle N, \sigma \rangle \cong \mathbb{Z}_p \rtimes \mathbb{Z}_p$ . By Proposition 3.33, there exists a valuation  $w$  on  $M := \text{Fix}(N, \sigma)$ , tamely branching at  $p$ , which we may assume to lie in  $H(M)$ . For  $p = 2$ , we additionally require that  $w$  does not have any proper coarsenings with real closed residue field. Let  $u$  denote the unique prolongation of  $w$  to  $F := \text{Fix } N$ . If  $\mathcal{O}_u \in H(F)$ , we see that  $u|_K$  is the desired valuation by **Henseling Down** (as indicated by the right diagram).

It remains to consider the case that  $u, v_F \in H_2(F)$  with  $\mathcal{O}_u \subsetneq \mathcal{O}_{v_F}$ . By definition,  $Fv_F$  is separably closed,  $\text{char } Fv_F \neq p$  (as  $Fu$  is a residue field of  $Fv_F$ ), and  $v_F F$  is not  $p$ -divisible (otherwise  $N = G_F$  would be trivial). Further,  $\mathcal{O}_u \subsetneq \mathcal{O}_{v_F}$  implies  $\mathcal{O}_w \subsetneq \mathcal{O}_{v_F|_M}$  (Fact 2.12), therefore,  $\mathcal{O}_{v_F|_M} \in H_1(M)$  since  $\mathcal{O}_w \in H(M)$ . By **Henseling Down**,  $u' := v_F|_L$  lies in  $H(L)$  and is tamely branching at  $p$  (since  $\text{char } Fv_F \neq p$ ,  $v_F F \neq pv_F F$ , and  $Mv_F|_M$  is not separably closed and  $\sqrt{-1} \in Lu'$ ). Finally, applying **Henseling Down** to  $L/K$  shows that  $v_F|_K = u'|_K$  is henselian and tamely branching at  $p$ .  $\square$



**Lemma 3.35** ([EP05, Lem. 5.4.6]). *Let  $(K, v)$  be a henselian valued field,  $v$  tamely branching at  $p$ . Then there exists a valuation ring in  $H(K)$  tamely branching at  $p$  as well. If  $p = 2$ , we can find a henselian valuation on  $K$ , tamely branching at 2, such that furthermore, no proper coarsening has real closed residue field.*